

Activity-3

Topics Covered: MVC & Design Pattern, Layout Management, Text Input Components, Choice Components,

1. Explain the MVC (Model-View-Controller) design pattern in Java Swing. Illustrate how it is implemented in Swing components like buttons.
2. Discuss the advantages of using MVC architecture in GUI applications. Explain how it improves modularity, maintainability, and scalability.
3. Design a GUI layout for a calculator form using appropriate layout managers. Justify your choice of layout managers.
4. A GUI freezes when resized. Explain how layout managers can solve this issue.
5. Design a student management system GUI using Swing. Explain the use of layout managers, input fields, menus, and advanced components.
6. Explain all text input components in Swing with examples and use cases. Include JTextField, JTextArea, JPasswordField, and JScrollPane.
7. Compare and analyze choice components in Swing: JCheckBox, JRadioButton, JComboBox, and JSlider with examples.
8. A desktop application needs multiple internal windows. Explain how JDesktopPane and JInternalFrame can be used to implement this.
9. Discuss how advanced components like JTable and JTree improve data visualization.
10. A music player application needs volume control and mode selection. Which Swing components are suitable?